Chapter 7 Production

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This chapter will cover several economics topics that have parallel applications in the C-Suite, though the same terminology isn't always used by the business community. A concordance of these terms is below.

Economics concept	**Practical business application**
Production technology	Operations research Standard operating procedures (SOP)
Isoquant	What are the different ways we can hit our production goals?
Marginal product	Optimizing production
Scale economics	Expansion plans

A common way to think about the operations of a business is as a chain of activities.[i] We have started at the end of that chain: the Chief Marketing Officer sells the products and services to customers. But as we move backwards along that chain, we see the organization must distribute those products and services after first producing them from a collection of inputs they acquire.

Figure 7-1

Every organization can think of what they do along this chain.

- For a dress manufacturer (physical product)
    - Inputs: fabric, assemblers (seamstresses and tailors), thread, patterns, sewing machines, and the building in which to make the dresses.
    - Production: the number, sizes, and styles of dresses
    - Distribution: warehousing, shipping (based on sales in next step)

    - Sales/service: omni-channel sales outlets to customers and consumers (own-retail/3rd party retailers, in-person and online/e-commerce), returns/repair of unwanted dresses
- For an accounting firm (service provider)
    - Inputs: Certified public accountants (CPAs), tax and bookkeeping software, computers, office space, partners to manage client relationships
    - Production: creating the income statements, balance sheets, tax returns, and other financial statements for clients, auditing clients' own financial statements
    - Distribution: Co-locating with client, sending materials electronically and by mail/carrier, submitting to state and federal agencies
    - Sales/service: Signing up new clients, advising clients on new offerings

Modern value chains can get very complicated and include partners (who could co-produce key inputs or products and services), complementors (who produce complementary products or elements that support the distribution), and platform providers (who could be a specific distribution channel, or a third-party logistics provider who enables the upstream portions of your value chain). But they always include these core elements: the organization needs to use inputs to produce something that they sell downstream, which they distribute to customers through various channels.

We've already seen in Chapter 2 a model that explains how the Input and Sales buckets work: the supply-demand market framework. But the middle two buckets – the production and the distribution – is a bit of a black box. *How* does the firm take those inputs and transform them into sellable products and services? This is what economists call ***production technology***. The computers and software that we typically think of as "technology" are actually inputs to the process for a firm; the (economic/production) technology is how those tools are used with all the other physical and intangible assets and resources the organization has to create the goods sold by the CMO.

7.1 Inputs

The examples of the dress maker and accounting firm highlight that there are multiple different types of inputs (and outputs) that a firm can produce. In fact, we have not even listed all the possible inputs (we're sure you could come up with a more exhaustive list in your head). But as economists, we simplify our production technology analysis down to two types of inputs: *labor* and *capital*. This partly is an artifact of when economics began as a discipline in the late 1700s. The economy was mostly agrarian, and you produced food by using laborers and implements (work animals, plows, scythes, etc.) purchased with capital. The labels have stuck because they align with the two types of inputs we still use today: *variable* and *fixed*.

Labor is a variable input because you can vary it in the short term. Capital is fixed because you cannot vary it in the short term. Think about the farm application: if you want to plant more crops, you can tell your workers to stay at the farm longer each day. If your crops didn't grow as planned, you can tell them to go home early at harvest time. The usage of the laborers can be varied in a short amount of time. But if you decide to stop planting crops on a given day, you can't get rid of the horses and oxen, or the plow – they are still at the farm. And if you want to plant more on a given day, you cannot go acquire those capital assets – it takes time for horses and oxen to grow to the point they can be useful in the field, and it takes time for the blacksmith to produce a new plow or tool.

In the long run, everything becomes variable. Five years from now, you can acquire more animals or tools, and if you are downsizing your farm, you can give those to other farmers (including the land itself, which is also capital). How much time must pass before we arrive at the "long run" depends on the industry. It's relatively easy to get a few computers and CPAs to expand an accounting firm in a few months, but to build a new factory to house the production of dresses could take at least 18 months or more. As economists, we can wave our hands and say, "As long as it takes for all the capital to be changed." In reality, that's the job of the COO to understand the time frames within which their organization operates.

The definition of labor and capital implies another way to think about the categorization. If we need to produce one more unit of output, do we need more of that input? For labor, the answer is "yes," while for capital it is "no." Think about farm production. If you want to plant or harvest another acre, do you need more time from the laborers? Certainly. Do you need another hoe? Another plow? Another ox? Not for one more acre. You might if you want to plant on another field, but even then, you might be able to use the same animals and implements you already have.

So we have two ways of thinking about our inputs: labor (L) and capital (K).[ii]

Figure 7-2

	Labor (L) **"Variable"**	**Capital (K)** **"Fixed"**
Varied immediately?	Yes	No
Need more to produce one more unit?	Yes	No

Let's think about how to apply this to our dress manufacturing and accounting firm examples above (the list below is not exhaustive).

Table 7-1

	Dress manufacturer	**Accounting firm**
Labor (variable inputs)	• Assemblers • Fabric • Thread • Electricity to run sewing machines	• Bookkeeper • Electricity to run the computers
Capital (fixed inputs)	• Sewing machines • Patterns • Factory	• Accounting software • Computers • Electricity to run the office (e.g., lights)

	• Electricity to run the factory • Supervisor • Chief Operating Officer	• Supervising partner • Managing partner

Notice that we have "people" in both Labor (Assemblers, Bookkeepers) and Capital (Supervisor, COO, Supervising Partner, Managing Partner). A human being is "labor" if their time is variable and more of it is needed to produce one more unit (i.e., dress, tax return). A human being is "capital" is their time is not varied in the short run (e.g., you don't fire the COO if you finish early for the day) and you don't need more of their time to produce one more unit (you don't hire an extra managing partner to file another tax return).

In economics, we think of small changes in output: one more dress, one more tax return. In the real world, the COO may not optimize to that specificity, but they do think about small changes to how much to produce and whether they can change the amount of inputs in a short period of time. An individual supervisor may be able to work overtime (more like labor), but it may take time to add another shift and another supervisor to the factory floor (more like capital). It matters less about whether you label each input accurately as labor or capital in the real world; what matters is that you recognize there are some inputs that can be varied in the short run to produce one (or a small number) or additional output, and others that cannot be adjusted and don't need to be changed to produce a bit more output.

Because the exact classification of capital and labor in real world example can get tricky, we'd like to simplify things for the remainder of the chapter. We'll use the example of a ditch digging business because at its core, it essentially has two inputs: the workers who dig (labor) and shovels (capital).

7.2 Production

There are three ways we can represent Production from Figure 7-1: a table, an equation, and an isoquant.

7.2.1 Table

The table form of production is like a spreadsheet. In Table 7-2, we have the amount of labor (the number of workers) in the rows and the amount of capital (the number of shovels) in the columns. The numbers in each cell are the total quantity (the number of meters you can dig in a given period of time – say, one day). We can extend this table for as many columns and rows as we want. In practice, we would extend it to account for the maximum number of workers and shovels we could use in any given day.

Table 7-2

		Capital				
		1	**2**	**3**	**4**	**5**
Labor	**1**	15	20	24	27	29
	2	40	46	51	55	58
	3	70	78	84	88	91
	4	110	120	128	134	138
	5	145	156	165	172	177
	6	172	182	190	196	201
	7	192	201	208	213	217
	8	201	208	213	217	220
	9	207	213	218	221	224
	10	207	214	219	223	225

Notice that there are three ways we can produce 201 meters of ditch:

- 8 workers and 1 shovel
- 7 workers and 2 shovels
- 6 workers and 5 shovels

How do we know these are the combinations we can use to dig that many meters per day? The COO will determine that by collecting lots of data through operations research. They will try many different combinations of capital and labor and count how many meters are completed at the end of the day. They record all this information in a database, which looks like Table 7-2 when it is printed out.

7.2.2 Function

A function is another word for a mathematical equation. We can represent the potential production output as a function such as:

$$Q = f(K,L)$$

where Q is the quantity of output (number of meters), f(•) means “function of the stuff in parentheses,” K is capital, and L is labor. The above notation is the most generic form of the ***production function*** and is the mathematical equation that tells the COO if they take a certain amount of capital (K) and labor (L), what quantity of output (Q) will they produce?

How does the COO develop this production function relationship? She doesn’t ask a GenAI bot or search engine to look it up: she tasks her data analytics team to take all the data in the table format (Table 7-2) and to use the team’s statistical prowess to estimate the equation that best fits the data. The team can then create a sheet at the front of the table workbook that includes the production function as a formula in a cell. The COO can enter various amounts of capital and labor into different cells on the front sheet, and the formula (production function) will reveal the total quantity produced.

The production function is a very useful concept to understand because it simplifies the decision the COO has to make when tasked with producing more output (Q). There are five ways the COO can produce more:

1. More capital [K]
2. More labor [L]
3. Better capital [K]
4. Better labor [L]
5. A better production function [f(•)]

Most underlying production technologies imply more output when you add more inputs, which explains the (1) and (2) above. More workers digging or more shovels leads to more meters. Numbers (3) and (4) should also be intuitive. If we have a backhoe instead of a shovel, we’ll be able to dig more. And if we train workers on the proper ergonomics for digging, they won’t tire as easily.

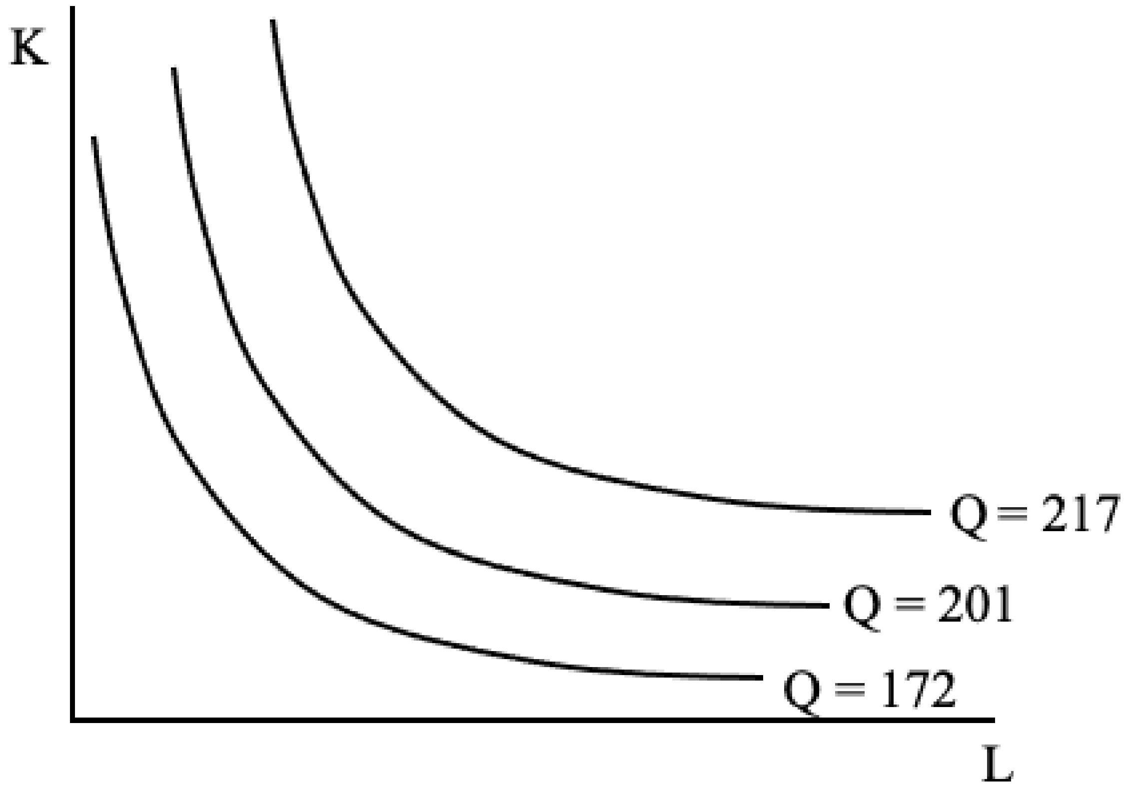
Number (5) says that if you figure out a better way of combining the workers and shovels together, you can get more output with the same inputs. For example, if instead of having one person moving along the trench to dig, they pass the shovel to the next person to start digging, you might be able to dig more in a given day. The final opportunity for production improvement is catnip for COOs: it's the equivalent of telling them, "Try everything you can to reorient the factory floor to improve output." The COO can then adjust the assembly line, which dock is used for receiving and which for shipping, where to store inventory, and whatever else she can think of – it's like playing with Legos to build a better mousetrap.

Our production function is very simple: the maximum quantity that can be produced with various combinations of capital and labor.[iii] But it can be tailored to represent any real-world production technology. We merely need to add more inputs inside the (•) part of our equation. Even though the actual production function for a real-world organization can be very complicated, it is still helpful to narrow down the operational decision to whether you can change capital or labor, improve capital or labor, or create a better way of doing things.

7.2.3 Isoquant

The final way of representing the production technology is with the isoquant, which means same (iso) amount (quant): what are the combinations of capital and labor that produce the same number of meters? We represent the isoquant on a graph with capital on the vertical axis and labor on the horizontal axis.

Figure 7-3



The shape of the isoquant reflects the underlying relationship of the production function. In economics, we often draw the isoquant as a curved line because it will make the optimization point easier to understand when we get to costs in Chapter 9, but the isoquant can be any downward-sloping shape. One extreme version is a straight line, which represents a one-for-one trade-off between inputs in the production function. In our ditch-digging example, if you had the option of using left-handed workers or right-handed workers, this would almost certainly be a straight-line production isoquant. If you produced 100 meters with 10 workers, it wouldn't matter if they were five lefties and five righties, or 10 of one type, or one and nine, or any other combination that added up to 10.

Another extreme isoquant is "L" shaped. This implies that you need a certain proportion of inputs to produce a given amount and any extra of either one is useless to total output. An example is producing a pair of shoes, where left-foot shoes are on the horizontal axis and right-foot shoes are on the vertical axis. These can be inputs if you outsource the manufacture of them to another manufacturer. If you have four left-foot shoes and four right-foot shoes, you can sell four pairs of shoes. But if you have four left-foot shoes and *five* right-foot shoes, you can *still* only sell four pairs. Similarly, 20 left-foot shoes and four right-foot shoes only yields four pairs of shoes.

These extremes are not the normal types of production trade-offs we see in the real world. Most production functions lie in between, so the isoquant is between a straight line and an "L" shape, or something like Figure 7-3.

7.2.4 Which one is correct?

We've shared three different ways to represent the production technology, so which one is the correct one? Which one do COO's use most often? All three! In fact, you might have already clued into this because we explained the production function is derived from the data in the table format, and the isoquant is just a graphical representation of the production function. All three are based on the same underlying data, and in the real world, this is what Enterprise Resource Management (ERM) software is. This software manages the operations of an organization, linking inventory and supply chain and production and shipping and distribution together to ensure that the maximal amount of output is distributed given the inputs and the processes of the firm. This software creates and houses massive databases (the "table" view), links the data to allow for forecasts of what can be produced (the underlying "production function" math), and often incorporates dashboards to allow the COO and her team to visualize the underlying relationships (the "isoquant").

Enterprise Resource Planning (ERP) software incorporates marketing and other functional areas into the planning process, while Human Capital Management (HCM) software aides in the assessment of variable and fixed "people" inputs to the production of the organization's output (essentially the production function for a services-based firm or non-profit). Regardless of the software provider or configuration, these tools are virtually ubiquitous in today's economy, and they all are methods to help the organization produce the goods and services it sells to customers by storing the data, calculating relationships between the data points, and visualizing them for the decision maker.

7.3 Marginal and average product

The marginal product and average product concepts help the Chief Operating Officer determine the trade-offs inherent in the optimization problem. We start with the marginal and average product of labor because in the short run, those variables inputs are the only ones the COO can change.

7.3.1 Marginal product of labor

The marginal product of labor (MPL) is the change in the total output when there is a change in the amount of labor used. We write it as:

$$\text{MPL} = \frac{dQ}{dL}$$

Using our ditch digging example, this is how many extra meters of ditch are dug when we use a small amount of extra worker time (usually whatever time period we are measuring labor usage). We calculate the MPL holding capital fixed (i.e., without changing the number of shovels used). We keep capital fixed partly because we're in the short run (so we can't change capital), but even in the long run we calculate MPL by holding capital fixed because if we change capital *and* labor at the same time, we're not sure whether the extra output was due to the labor, the capital, or both. By holding capital fixed and changing only labor, we know that any change in the quantity produced must be due to the change in labor.

In Table 7-3, we've used the underlying production data from Table 7-2 where capital equals one (column C in the previous column is the same as the "Q" column below). We've added a column for the MPL, which is the difference between Q for that row minus the Q from the row above. For example, one worker and one shovel produce 15 meters, but two workers and one shovel produce 40 meters. The additional worker has added 25 additional meters, which is the marginal product of labor.

Table 7-3

Labor	Capital	Q	MPL
0	1	0	-

1	1	15	15
2	1	40	25
3	1	70	30
4	1	110	40
5	1	145	35
6	1	172	27
7	1	192	20
8	1	201	9
9	1	207	6
10	1	207	0

Why does the second worker add more meters than the first worker? If one worker digs 15 meters, then shouldn't two workers dig 30 meters? And since they have only one shovel, how can the second worker possibly even dig 15 meters, since they don't have a shovel?!?

It is *not* because the second worker is somehow "better" or "stronger" than the first. Remember that the production function is the maximum amount you can produce with a given amount of inputs, so with one shovel and one worker, the production function is implicitly assuming you're using the "best" worker first. We would believe that in the real world, the COO of the ditch digging company would figure out over time who the best workers were and make them the initial ones on call to come and dig each day.

There are two primary reasons why the marginal product of labor increases, and they are generalized across other industries and production functions. They are both variations on opportunities provided by cooperation ("operating" together).

7.3.1.1 Collaboration

The simplest reason the two workers can be collectively more productive is that they can rest while the other is working and alternate who is working. Think about one worker digging by themselves. In the first hour of the day, they are refreshed and energetic. They start to slow down in the second hour, and the third, and by the end of the day, they can barely lift the shovel to move dirt around. If there are two workers, however, then the first one can rest in hour two while the second worker digs. Then in hour three, the first one resumes digging and the second one rests. By the eighth hour of the day, the second worker is digging with the energy of the fourth hour of digging, but with even more energy than a sole digger in hour four because the second worker has had three hours of rest after digging for an hour.

We can get more total amount of work done – and output – while still having only one person digging at any one time, simply by alternating who is working. This would be true of knowledge work, too: the rest allows individuals to recoup their energy and focus and be more productive later in the day. The productivity gets even better if we allow the two workers to be digging at the same time during some of the hours and alternating their rest at different times. The ditch continually gets longer while the workers are more energized later in the day.

The second reason collaboration can be beneficial is the potential to create better engagement and emotional support because the workers have someone else to dig with. Recall a time when you had to work by yourself – it can get tedious very fast. If you have someone else to share the time with, it can lift your spirits and make you more productive at the end of the day.

In addition, having another person to work with could create innovative ways of working that increase output further. This is not a change to marginal product of labor because we are changing the entire production function: the f(•) changes. But over time, this is another benefit of collaboration.

7.3.1.2 Specialization

The other reason that two workers can be more productive than doubling the output of the first worker is that each individual can specialize in a particular task that needs to be accomplished. Ditch-digging does not seem complex, but the worker digging by themselves needs to loosen the dirt, put a pile on the blade of the shovel, heave the dirt outside the ditch, and carry the dirt away for disposal (because a ditch isn't much use if it's blocked by the dirt that was excavated). At a minimum, when the worker is hauling the dirt away from the dig site, no additional meters are added to the ditch. But if we have two workers, one can haul dirt while the other shovels and heaves the dirt out of the ditch. Or one person can loosen dirt while the other heaves it out with the shovel.

They can also take turns shoveling and hauling dirt (our collaboration point above). They can take turns while also giving each other breaks (especially if it takes longer to loosen and excavate the dirt than it does to haul it away).

Specialization and collaboration can combine to make the second worker add more additional meters to the ditch than the first laborer did.

7.3.2 Law of Diminishing Returns

If we return to Table 7-3, we can see the third worker adds an incremental 30 meters, while the fourth increases the length of the ditch by 40 extra meters. But notice that the fifth worker only adds 35 meters when they join the dig site, while the sixth adds an incremental 27, the seventh worker adds 20, and each subsequent digger added to the site adds less than 10 meters. The length of the ditch gets longer with each additional worker up through nine, but the rate at which it gets longer decreases. This is called the Law of Diminishing Returns. Eventually, extra workers add to the total output of production at a decreasing rate: the marginal product of labor decreases. It doesn't have to be negative (though it could), and it doesn't have to be zero (though it could, as we have with the tenth worker in Table 7-3). But the marginal product of labor does eventually decrease as more workers are added.

One reason this occurs is that workers get in each other's way. Visualize the dig site. There are only so many workers who can fit inside the ditch before they start bumping into each other and hitting each other as they swing their arms back to shovel the dirt. Whether they politely discuss who has the right to dig at that spot, or they throw down their shovels and start arguing, they are not excavating any ground, and the ditch does not get longer.

The same happens on a factory floor, or in an office space. Even the most intricately arranged cubicles will eventually run out of room for knowledge workers to sit and work at their computers. And the assembly line on the factory floor cannot accommodate more than one worker at a spot.

The second reason why diminishing returns occur is that there are limits to specialization. By the time the tenth worker arrives at the dig site, there are enough people to dig, haul away dirt, bring drinking water to slake people's thirst, and to provide rest for each other. The tenth worker will circle around the site asking if they can help out, continuing to get replies of, "Thanks, but we've got it" until they decide to simply sit down and wait for the day to end.

7.3.3 Average Product of Labor

The other key metric related to each additional worker is the impact on the average product of labor (APL), or the average output each worker produces. This is also called "labor productivity," and is measured simply as:

$$\text{APL} = \frac{Q}{L}$$

This metric is used by the Chief Operating Officer to compare the average output for each worker across different factories, shifts within a factory, back-office support locations, warehouses, geographies, or any combination of the above. The particular categories the COO chooses depends on the way in which the organization structures itself.

The APL for our ditch-digging example is shown in Table 7-4 as the last column. It is the third column divided by the first.

Table 7-4

Labor	Capital	Q	MPL	APL
0	1	0	-	-
1	1	15	15	15.0
2	1	40	25	20.0
3	1	70	30	23.3
4	1	110	40	27.5
5	1	145	35	29.0
6	1	172	27	28.7
7	1	192	20	27.4
8	1	201	9	25.1
9	1	207	6	23.0
10	1	207	0	20.7

The average product of labor is related to the marginal product of labor in the following way:

- If the MPL > APL, then the average product of labor is increasing
- If the MPL < APL, then the average product of labor is decreasing

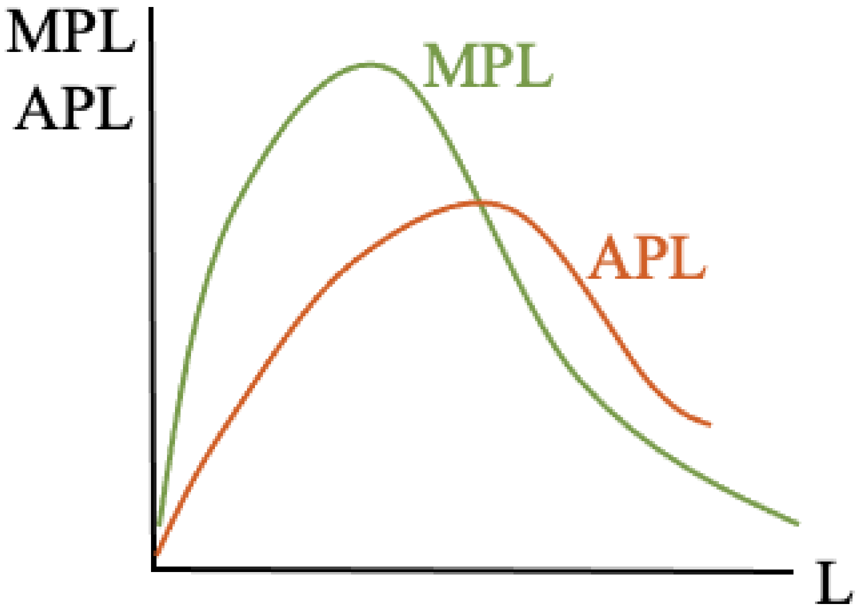
We can see this in Table 7-4. Up through the fifth worker, MPL is greater than APL. But starting with the sixth worker, the MPL is less than the APL. And the average product of labor starts decreasing with the sixth worker.

There is no economic intuition for this: it's just math. If the last worker adds more than the average, then you're adding the average – plus you have some left over the you can spread across all the workers. Therefore, the average increases. Similarly, if the next worker adds less than the current average, then to you have to "borrow" from all the others to try and get back up to the average, which results in the average falling overall.

7.3.4 Visualizing the MPL curve

The marginal product of labor increases and then decreases (law of diminishing returns), which means when we graph it – where the MPL is the vertical axis and the quantity of labor (L) is the horizontal axis – it appears hump-shaped, as the green curve in Figure 7-4.

Figure 7-4



We can overlay the APL on the same graph, with the only change being the vertical axis measures the average product of labor for each level of L. This curve is also hump-shaped because of the "math" property we described above: if the MPL is greater than the APL (MPL lies above APL on the graph), then the APL is increasing, and APL falls if MPL lies below it. (This also means that the MPL intersects the APL at the maximum point on the APL curve, as drawn above.)

We can also use this graph to think about how the marginal product of labor changes as we add more capital. Because each worker now has more capital to work with, the marginal productivity of that worker will be greater. That doesn't necessarily mean the curve shifts up parallel – as drawn in Figure 7-5, the curve shifts up by a greater amount when the amount of labor is greater (on the right side of the curve).

Figure 7-5

We can relate that to our ditch-digging example. On the left side of the curves, we have low levels of labor – say one digger. On the right side, we have more – say 10 workers. Imagine MPL_1 is drawn with one shovel, while MPL_2 represents the marginal product of labor when there are two shovels. With one worker, that worker will get a bit more productive with a second shovel (e.g., they can use a shovel in each hand to hack away and loosen the dirt, then use one of the shovels to heave it out of the ditch). But the tenth worker will become *more* productive when there is a second shovel added (e.g., they may not be able to use the shovel themselves, but there is now one more worker digging with a shovel, which means more dirt is excavated and there is more for the tenth worker to do).

How do we know the exact shape of these curves and how much the curve will shift when we add more capital? Operations research. The COO conducts tests and experiments and builds the database that informs them of all the trade-offs involved when the production process is altered.

7.3.5 Marginal and average product of capital

That last part might have left you wondering: I thought we were keeping capital fixed in the short run? That's correct, but in the long run the COO can change the level of capital the firm uses. In our simple example, they can get more shovels and bring them to the dig site.

In fact, we can create the same metrics we did for labor. The marginal product of capital, or the change in the total output when we change the amount of capital by a small amount, is:

$$\text{MPK} = \frac{dQ}{dK}$$

And the average product of capital, of the amount produced by each unit of capital, on average, is:

$$\text{APK} = \frac{Q}{K}$$

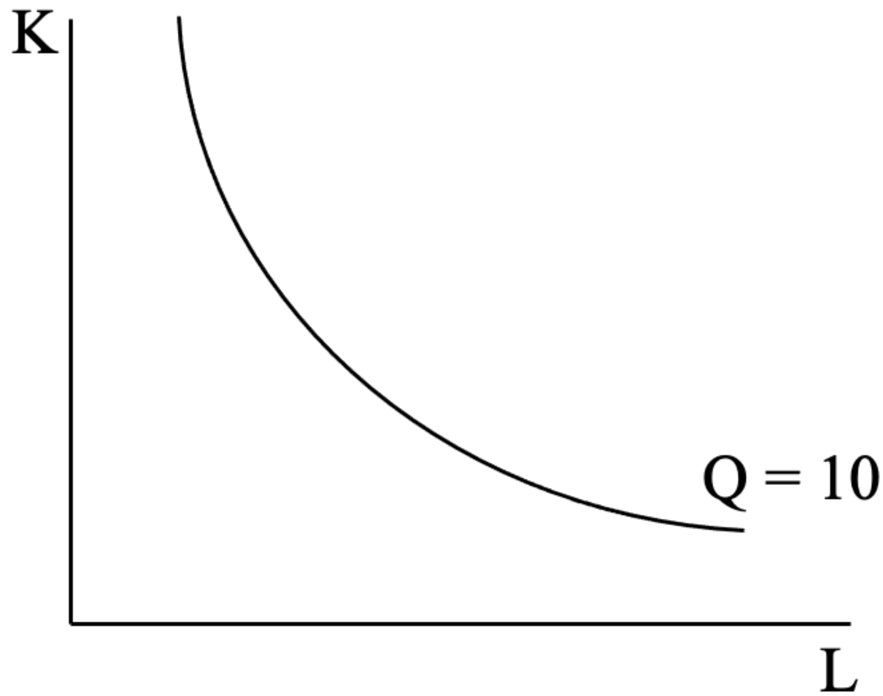
The MPK exhibits diminishing returns to capital (if you throw more and more shovels at the dig site, eventually they just sit there and add nothing to the excavation efforts), and if the MPK is greater than the APK, the APK is increasing, and if MPK is less than APK, the average product of capital is decreasing (again, because of math).

The marginal product of capital is an important concept for thinking about the trade-off of capital and labor, which is where we turn next.

7.4 Marginal Rate of Technical Substitution

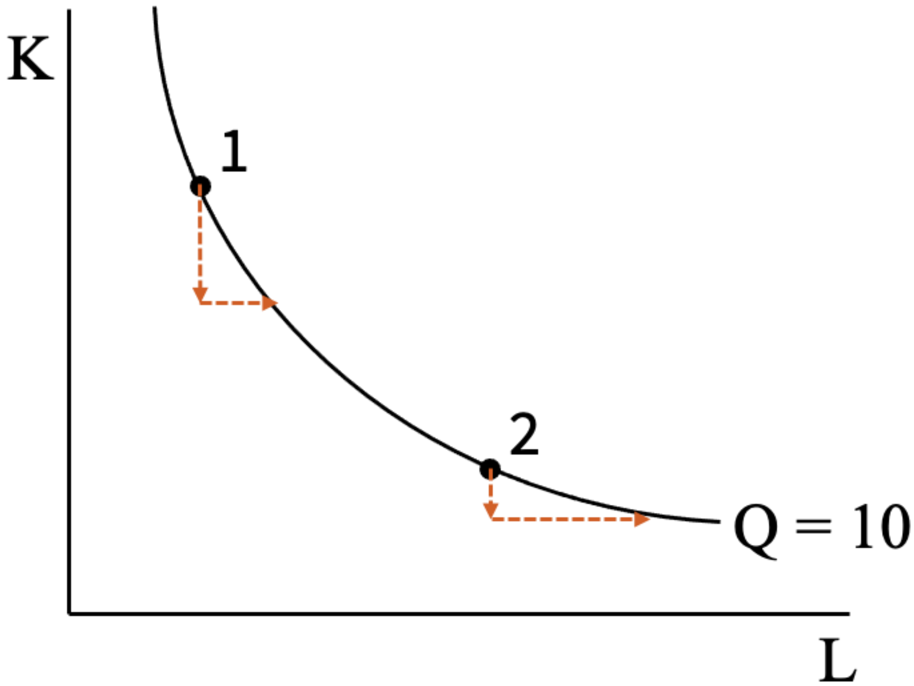
The Chief Operating Officer can switch using capital for labor, or vice versa, in the long run. The COO would want to know all the possible combinations of capital and labor that could be used to produce a certain amount of output – their production target. We represent this trade-off as the isoquant: "iso" means equal, and "quant" means amount, so as seen in Figure 7-6, it's the combinations of capital (K on the vertical axis, the shovels) and labor (L on the horizontal axis, the workers) that produce equal amounts of output (or meters of ditch, in our example). Here, we've drawn the isoquant for producing 10 meters of ditch.

Figure 7-6



The slope of the isoquant represents the trade-off of capital and labor at any one point on the curve, and this slope changes as we move along it. In Figure 7-7 we've replicated the curve, but we've added two points to highlight how the slope of the isoquant changes and is dependent on how productive each input is.

Figure 7-7



- At point 1 on the upper left, we have a lot of shovels (K) and very few workers (L). Because there are a lot of shovels, each individual one is not very productive. In fact, some of them might not be used, so the marginal product of capital – the change in output when we change the number of shovels – is very small. We can give up a lot of shovels (travel down the dotted arrow) in order to add one additional worker and result in the same total amount of meters of ditch (Q). The workers are very productive because we have a lot of shovels lying around, and adding one more worker means they'll immediately grab a shovel and start excavating.
- On the other hand, at point 2, we have a lot of workers (L) and very few shovels (K). Each shovel is very useful because there are multiple workers who'd want to use each one. But the excess number of diggers relative to shovels means they are not individually adding much to total output. If the COO gave up just a small number of shovels (travel down the dotted arrow from point 2), if they gave up just a few shovels, they would have to add a large number of workers to compensate and get back to the same 10 meters of ditch.

The relative trade-off between capital and labor depends on how productive each incremental unit is, which we've seen is the measure of marginal product of labor (MPL) and the marginal product of capital (MPK).

We call this slope of the isoquant the marginal rate of technical substitution, or MRTS, because it tells us the *rate* at which we *substitute* capital for labor (or vice versa) in the *technology* production function for small (*marginal*) changes in input levels. It depends on the MPL and the MPK, or how much output changes when we change each input by a small (marginal) amount. Mathematically, it is measured as:

$$\text{MRTS} = -\frac{MPL}{MPK}$$

(We'll use this ratio in Chapter 9; if you want to see how it's derived, see the box below.)

MATH

The MRTS is the slope of the isoquant, which we can find using the production function.

$$Q = f(K,L)$$

Taking the total derivative of this equation tells us how output, Q, changes if we change either capital (K), labor (L), or both.

$$d\text{Q} = \frac{d[f(K,L)]}{dL} * d\text{L} + \frac{d[f(K,L)]}{dK} * d\text{K}$$

The first term is the change in output when we change labor, times how much we change labor (which is a measure of the change in output). The second term is the change in output when we change capital, times how much we change capital (which is a measure of the change in output). So we have the change in output due to adjustments to labor and the change in output because we adjusted capital, or a total change in output (the *d*Q on the left).

If we want to calculate the slope of the isoquant, that means we want to keep output constant, so the change in output equals 0.

$$dQ = 0$$

$$0 = \frac{d[f(K,L)]}{dL} * dL + \frac{d[f(K,L)]}{dK} * dK$$

Rearrange by subtracting $\frac{d[f(K,L)]}{dL}$*dL from both sides yields

$$-\frac{d[f(K,L)]}{dL} * dL = \frac{d[f(K,L)]}{dK} * dK$$

We can simplify this by replacing f(K,L) with Q, since that's the production function:

$$-\frac{dQ}{dL} * dL = \frac{dQ}{dK} * dK$$

We can now substitute MPL for $\frac{dQ}{dL}$ and MPK for $\frac{dQ}{dK}$, since those are the definitions of each marginal product term:

$$-\text{MPL} * dL = \text{MPK} * dK$$

We want to calculate the slope of the isoquant, or $\frac{dK}{dL}$, or $\frac{rise}{run}$, so we divide each side by dL:

$$-\text{MPL} = \frac{dK}{dL} * \text{MPK}$$

Now, we divide each side by MPK, so we have:

$$-\frac{MPL}{MPK} = \frac{dK}{dL}$$

Therefore, the slope of the isoquant is $-\frac{MPL}{MPK}$.

Look at point 1 in Figure 7-7. We have a lot of capital and very little labor, so adding an incremental unit of capital will have very little effect on output (MPK is small), while adding another worker will have a big impact on output (MPL is large). Therefore the slope is a large number divided by a small number, or a large MRTS. A steep slope.

Similarly, at point 2, we have a lot of labor and very little capital. Each incremental shovel will add a lot of meters to the ditch (MPK is large), while another worker will barely extend the trench (MPL is small). A small number divided by a large number is small, or a small MRTS. A flat slope.

7.5 Returns to scale

The Chief Operating Officer also needs to understand how they can expand operations if needed. The expansion plans will reply on increasing both capital and labor in the long run, and the COO needs to think about the effect of an increase in *both* capital and labor on output. (This is different from the MPL and MPK concept, where we changed just one input at a time.)

Let's take the example of a restaurant owner. She starts with a small kitchen, with one stovetop, one oven, one set of pots and pans, three feet of counter space, etc. She can produce a certain number of meals each evening at the restaurant (20, just to give it a number).

What if she increases everything by a factor of five (5): five stovetops, five sets of pots and pans, five ovens, five employees, fifteen feet of counter space, etc. It's quite likely that she could produce more than five times the number of meals; say she can serve 120 each night. She will be able to serve more customers because when it was just her in a small kitchen, she had to chop the ingredients, cook everything, plate each dish, and clean the pots and pans (and dinner plates). While she was chopping, plating, and cleaning, she was not preparing meals. But with a larger kitchen and staff, specialization allows the team to continually be preparing meals – there are no breaks for chopping and cleaning: one person chops, one person cooks, one person plates, one person cleans, and one person pitches in where needed.

Now let's think about increasing the size of the restaurant's kitchen by a factor of five *again*. Twenty-five employees, with 25 set of pots and pans, 75 feet of counter space (almost the length of a basketball court), 25 ovens, and 25 stovetops. In this case, she can produce more meals, but probably less than 600: fewer than five times the amount she had with the medium sized kitchen. It's not efficient to have that big of a kitchen because specialization means we have one chef just to cook the Brussels sprouts, and if no one orders then on a given night, that person stands around and does nothing. They don't add to the production of meals that night. And even if everyone does cook, or chop, or clean their specialty each night, the time it takes to walk from one end of the basketball-court-sized kitchen to the other to deliver the Brussels sprouts, and the chicken, and the potatoes to the area where the plating occurs means there is a lot of wasted time that is not productively used for preparing meals.

Small restaurants have too much being done by the small number of workers and we lose the opportunity for continual sequencing of activities. But really large restaurants have workers doing nothing or spending a lot of time walking around the space – both of which don't result in more completed meals.

This sizing issue is true for every industry. In economics, we think about what happens to output if all the inputs are increased by the same amount. In our restaurant example, we increased everything by a factor of five, but we simplify the calculations by imagining what happens to output if we double all the inputs. There are three possible results:

1. Increasing returns to scale: the total amount of output more than doubles
2. Constant returns to scale: the total amount of output exactly doubles
3. Decreasing returns to scale: the total amount of output less than doubles.

Our restaurant owner saw increasing returns to scale when growing from a small kitchen to a medium-sized kitchen, and she saw decreasing returns to scale when expanding to a large kitchen. This is generally true for most industries: at small sizes there are increasing returns to scaling up (getting bigger), and there are eventually decreasing returns to scale as expansion continues. If this weren't true, there would be one super large coffee shop that would optimally serve an entire city. Companies in a certain sector tend to be of similar sizes:

- Sandwich shops are small retail establishments

- T-shirt printing shops are the size of a large garage
- Professional services firm offices might have a hundred associates and partners combined (and not 1,000 in the same building)
- Amazon warehouses are the size of multiple football fields (but not the size of a large town)
- Semiconductor fabs are small rooms enclosed in buildings of roughly the same size (because it's harder to keep a big room clean from dust than it is a small room)

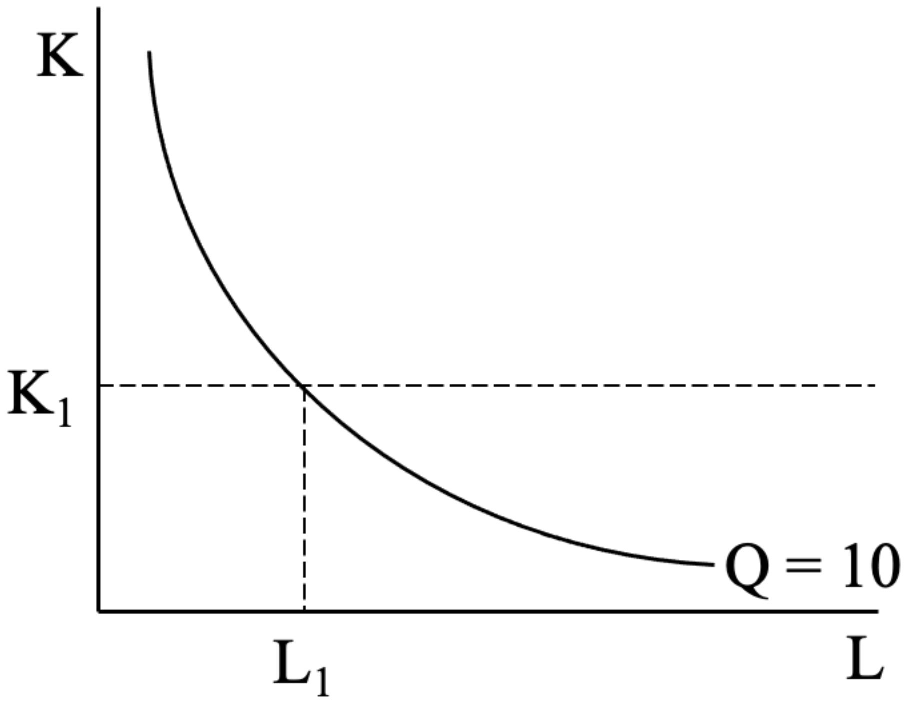
At some point, it is more efficient to create another facility of the exact same size (constant returns to scale), than it is to continually expand the size of an existing facility.

7.6 Short- vs. long-run choices

So what choices should the Chief Operating Officer make? We've hinted that there is an optimal size of each facility or office, but how should the COO think about the trade-off between capital and labor?

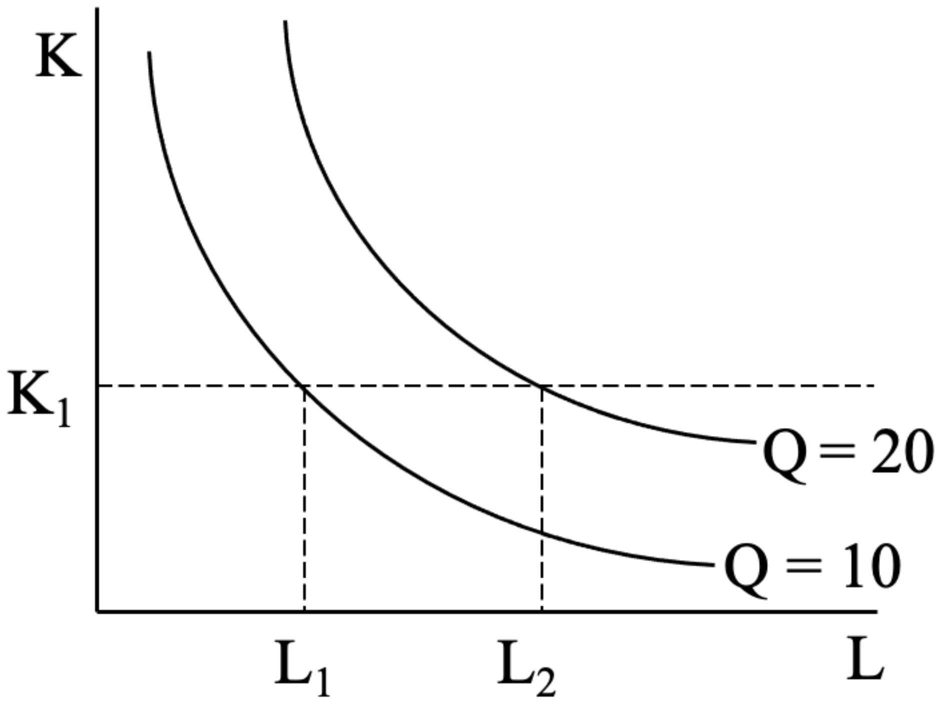
Let's say our ditch-digging company is producing 10 meters every day, and they start with K_1 units of capital. As seen in Figure 7-8, the COO needs to hire L_1 units of labor to produce the 10 meters each day. That's because it is the point on the isoquant where the COO is using K_1 units of capital.

Figure 7-8



But let's imagine the COO now needs to produce 20 meters each day, as seen in Figure 7-9. Why might that occur? One simple reason is the Chief Marketing Officer just sold a contract with a new customer who wants 10 meters of trench dug daily. Or an existing customer increased their order size. Or the CMO has finally expanded the sales operations in a new country. Regardless of the reason, the production target has increased, so what is the COO to do in the short run? There is only one option: increase the amount of labor used to L_2, since this is the only place where the COO can produce 20 meters with the existing K_1 shovels.

Figure 7-9



What should the COO do in the long run, when the number of shovels can be changed? Should they continue to produce at (K_1, L_2)? Should they move up the isoquant to the northwest and use more capital and less labor? Or should they move down to the southeast and use even *more* labor and give up some shovels?

Your instinct might be to move up the isoquant and use more capital, but what if I told you that extra shovels cost $1,000, while we could hire an additional worker for $10 an hour? Would to want more shovels and fewer workers? Conversely, what if I informed you that shovels could be purchased for $5, while a worker would cost $35 per hour? Now what would you do?

The answer depends on the cost of each of the inputs, and how much budget the COO can spend on the necessary inputs. But that's not the Chief Operating Officer's domain: that's the responsibility of the Chief Financial Officer.

Before we shift to the CFO, though, we need to address the behavioral biases that affect the COO, and in the next chapter talk about the impact of asymmetric information on the Chief Operating Officer's function.

7.7 Behavioral biases

There are three behavioral biases that can affect the COO, and these diverge from those we saw in the CMO's domain because they affect how the Chief Operating Officer makes decisions, not the suppliers (or customers, as we saw with the Chief Marketing Officer section).

7.7.1 Status quo bias

For all the talk about change being good, and people claiming they want to experience variety and change, human beings have a tendency to like things to remain the same. This is the status quo bias, and it's related to loss aversion (we don't like losing things we have) and the endowment effect (which we saw in the CMO domain of Chapter 3.

The status quo bias is especially challenging when it impacts the COO's decision making. You want the COO at your organization to crave experimentation – in fact, really good COO's would *love* to have the opportunity to take two weeks off to go redesign the factory floor, or to create new routing maps for logistics delivery. The COO who instead says, "This is working fine, and if we change, who knows how bad it's going to get?" is suffering from the status quo bias. Better ways of operating – process innovation – will suffer in the name of keeping things the same.

7.7.2 Law of the instrument

This bias involves overreliance on familiar tools. It is commonly referred to as "If all you have is a hammer, everything looks like a nail." This bias can rear its head when a new Chief Operating Officer falls back on an old bag of tricks: someone who came up through the ranks as a supply chain expert might have a tendency to "fix" operational problems by streamlining the supply chain. Or someone who was responsible for downstream logistics would address warehousing as a problem instead of looking into issues on the factory floor.

This highlights the importance of rotational programs to expose individuals in any functional area to various roles and responsibilities. When those individuals rise up the ranks and become C-suite leaders, they'll be less likely to pull out their hammer each and every time.

7.7.3 Not invented here

The final bias relates to the resistance to using knowledge or processes developed outside of the organization. Be wary of a Chief Operating Officer who claims, "That may work for our competitors, but it would never work with us." Each organization is different, with a unique combination of assets, resources, capabilities, and competencies. But that does not imply that there is nothing to learn from how others function within your sector – or in other industries.

The COO should always be scanning other competitors, geographies, and industries for ideas to adapt to the situation within their organization. Just because an idea wasn't invented inside your "black box" does not mean it cannot be a source of inspiration.

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[i] Porter, M. E. The Competitive Advantage: Creating and Sustaining Superior Performance. NY: Free Press, 1985. (Republished with a new introduction, 1998.)

[ii] We use "K" for capital because "C" was already used for "cost."

[iii] We assume the production function is the maximum quantity possible because we can always produce nothing – the workers can sit on the side of the ditch and do nothing – but that's not interesting for the COO. Remember from Chapter 1 that a core economic principle is optimization, which implies producing the most possible with each combination of inputs.